

Customer Churn Analysis

Simple English Interview Explanation

1. 1-Minute Project Explanation

“My project is Customer Churn Analysis. The main objective was to understand why customers are leaving the company and which customer groups have higher churn.”

I used **Python, Pandas, NumPy, Matplotlib and Seaborn** for the analysis.

First, I loaded the customer churn dataset and explored the data using functions like `shape`, `head()` and `value_counts()`. Then I checked for missing values and cleaned the data. For example, I converted the `TotalCharges` column into numeric format and handled the missing values.

After cleaning the data, I performed **Exploratory Data Analysis**. I used count plots, distribution plots, KDE plots, correlation analysis and a heatmap to understand the relationship between different customer attributes and churn.

I also converted the categorical variables into dummy variables and converted Churn into a binary variable, where **0 means No Churn and 1 means Churn**.

From the analysis, I found that **month-to-month customers, customers with higher monthly charges, customers in their first year, Fiber optic customers, and customers without Online Security or Tech Support** showed higher churn in this dataset. Customers with long-term contracts and longer tenure showed lower churn.

So, the main purpose of my project was to identify these churn patterns so that the company can further investigate these customer segments and develop appropriate retention strategies.”

2. If the Interviewer Asks: “What Exactly Did You Do?”

“I mainly worked on the **data cleaning and exploratory data analysis** part. I loaded the dataset using Pandas, checked the structure and missing values, cleaned the `TotalCharges` column, converted categorical variables into dummy variables, and converted the Churn column into 0 and 1. Then I created different visualizations to compare churn across customer characteristics and used correlation analysis and a heatmap to identify important relationships.”

3. Explain the Project Step-by-Step

Step 1 — Objective

The objective was to understand customer churn and identify the factors associated with customers leaving the company.

Step 2 — Dataset

I worked with a customer churn dataset containing customer demographic information, services, contract details, payment method, monthly charges, total charges and churn information.

Step 3 — Data Exploration

First, I explored the dataset using Pandas. I checked the number of rows and columns, looked at the data, checked the target variable and examined the distribution of churn.

Step 4 — Data Cleaning

I checked for missing values. I found an issue with the TotalCharges column because it was not completely numeric. I converted it to numeric using `pd.to_numeric()` and handled the resulting missing values.

Step 5 — Data Transformation

After cleaning, I converted the categorical columns into dummy variables so that I could perform numerical analysis and correlation analysis. I also converted Churn into 0 and 1.

Step 6 — EDA

I performed univariate and bivariate analysis. I created count plots to compare different customer categories and churn, and analyzed Monthly Charges, Total Charges, Tenure, Internet Service, Online Security and Tech Support.

Step 7 — Correlation Analysis

I used correlation analysis and a heatmap to understand which variables had stronger relationships with churn.

Step 8 — Findings

The major patterns were higher churn among month-to-month customers, customers with higher monthly charges, first-year customers, Fiber optic customers, and customers without Online Security or Tech Support. Longer-tenure and long-term contract customers showed lower churn.

4. Tools Used

Tool	Purpose
Python	Main programming language used for the analysis.
Pandas	Data loading, cleaning, transformation and analysis.
NumPy	Numerical operations and creating the binary churn variable.
Matplotlib	Data visualization.
Seaborn	Statistical plots and visual analysis.
Jupyter Notebook	Environment used to perform and document the analysis.

5. Common Interview Questions & Simple Answers

What is the objective of your project?

"The objective was to understand customer churn and identify the customer characteristics associated with customers leaving the company."

Why did you use EDA?

"I used EDA to understand the dataset, identify missing values, find patterns, and understand which variables were associated with churn before doing any further modeling."

What is churn?

"Churn means a customer has stopped using the company's service or has left the company. In my project, Churn = 1 means the customer churned and Churn = 0 means the customer did not churn."

What was the most important finding?

"Month-to-month customers showed higher churn compared with long-term contract customers. Higher monthly charges and lack of Online Security or Tech Support were also associated with higher churn."

Why did you convert categorical variables into dummy variables?

"Dummy variables convert categories into numerical 0/1 values, which makes them suitable for numerical analysis and correlation calculations."

What did you do with TotalCharges?

"I converted TotalCharges to numeric because some values were not recognized as numeric, then handled the resulting missing values."

What would you do next?

"I would build a churn prediction model, evaluate it using suitable metrics, identify the most important predictors, and then use the results to support customer retention analysis."

6. Easy Way to Remember

**Load → Explore → Clean → Transform → Visualize → Analyze →
Conclude**

Whenever the interviewer says “Explain your project”, follow these seven steps and speak naturally rather than memorizing every sentence.